AMP8611 - Compact Integrated Stereo Power Amplifier

V.2.6

Amplifier combining a 2-channel input selector, high-performance PGA2311 digital volume control and LM3886 power stages.

Key Specifications:

- Input resistance: 10kΩ;
- Nominal input voltage: 0.85Vrms;
- Maximum input voltage: 2.5Vrms;
- Input Channels isolation: 95dB;
- Maximum output power: 2 x 45W RMS (1kHz sinusoidal, 4ohm load);
- Bandwidth (-3dB): 20Hz – 90kHz;
- Volume control: -80 to +12dB;
- Dimensions: 240 x 265 x 65 mm (80 mm including feet);
- Weight: 4.0 kg.

Full documentation (description PDF, schematic, PCB layout, Gerber files, BOM and firmware) is available in the repository https://github.com/dmsd-lab/AMP8611.

Preamplifier board PREAMP2311

Preamplifier based on PGA2311 - high-performance stereo volume control for professional and high-end consumer systems.

Input channel selector based on 74HCT4052 and OPA1678 in an ultra-low-distortion, series-shunt current-mode configuration.

Bipolar power supply based on LM317L and LM337L linear regulators.

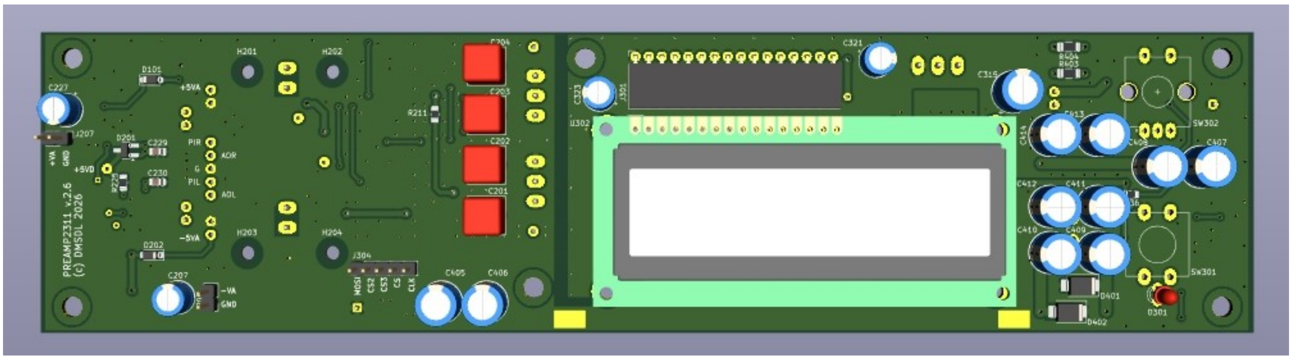
STM32G070 MCU, 2x16 LCD (with lowering backlight brightness after user inactivity)and rotary encoder with button for user interface (input channel selection, volume control).

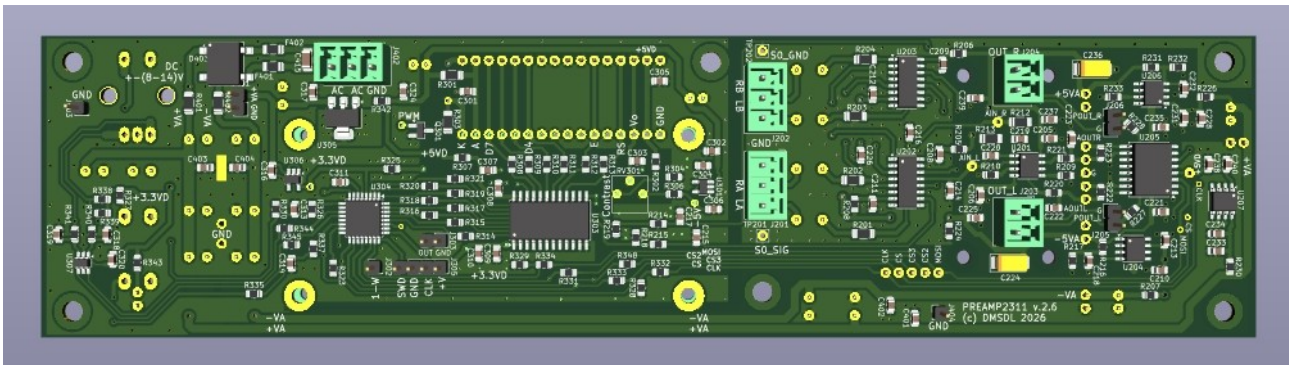
Supports integration of controlled forced cooling and a digital tone control / Bluetooth module.

Key Specifications:

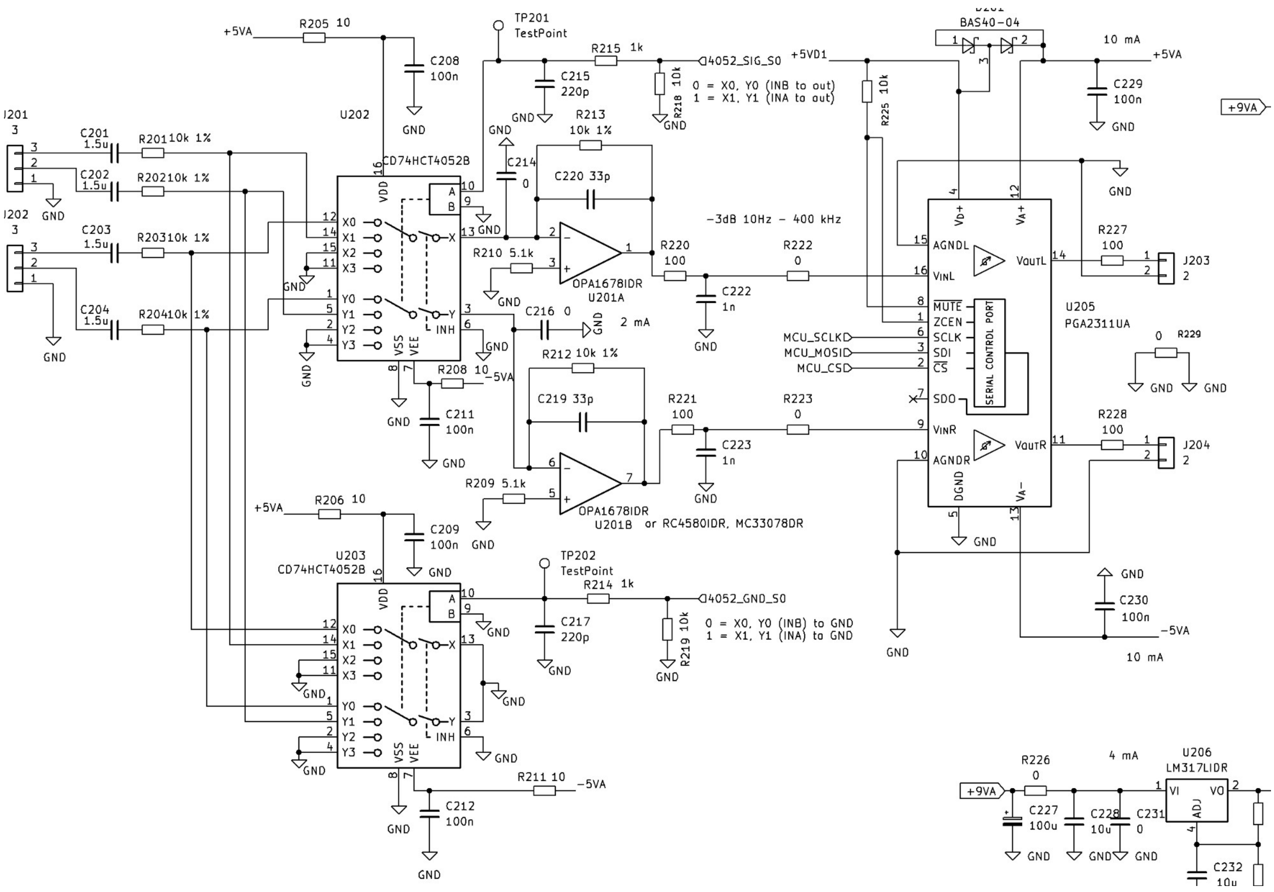
- Power supply: 2x(7 – 11V) AC or +-(8 – 14)V DC, 3W;
- Input resistance: 10kΩ;
- Maximum input voltage: 2.5Vrms;
- Output offset voltage (gain 0dB), maximum: +-2.5mV;
- Bandwidth (-3dB): 10Hz – 350kHz;
- Channel isolation: 95dB;
- Volume control: -80 to +12dB (−95.5 dB to +31.5 dB firmware limited);
- PCB dimensions: 234 x 58mm.

Firmware is developed with STM32CubeIDE.
The volume control step configuration is as follows:
−80 to −40dB - 5dB steps;
−40 to 0dB - 2dB steps;
0 to +12dB - 0.5dB steps.





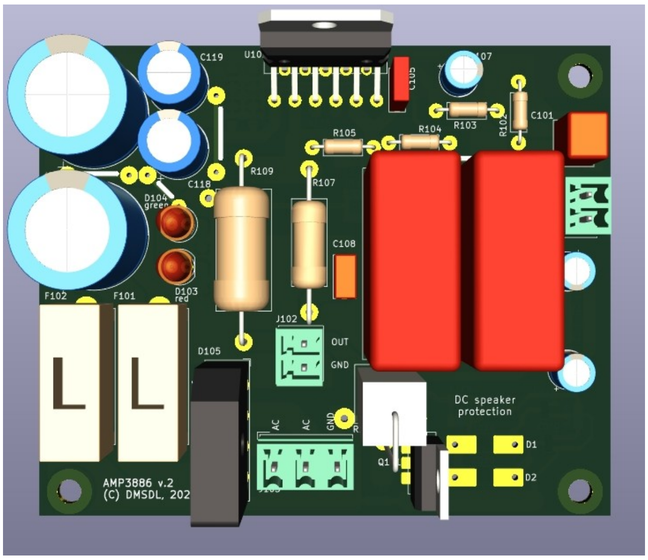
PREAMP2311 signal stage schematic

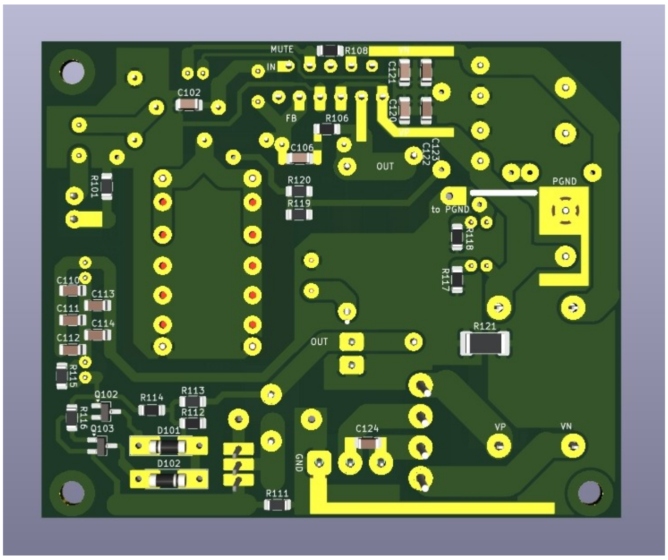


Power amplifier board AMP3886

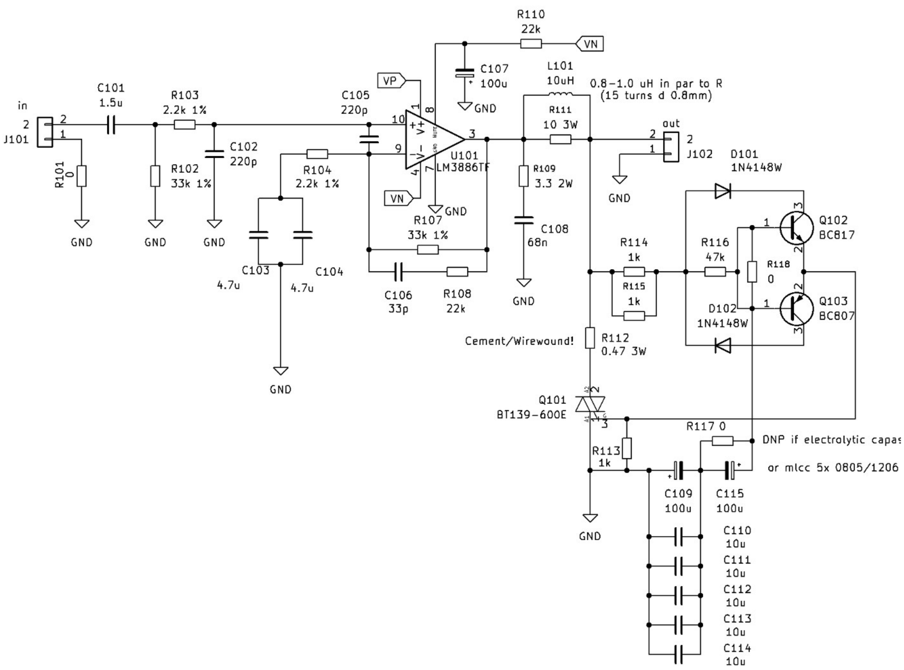
Power amplifier stage based on LM3886TF.
Single-layer DIY-optimized PCB.
Key Specifications:

Power supply: 2x20V AC (for 4Ω load)/2x25V AC (for 8Ω load), 100W;
Input resistance: 33kΩ;
Nominal input voltage: 0.85Vrms;
Gain: 24dB;
Maximum output power: 45W RMS (1kHz sinusoidal, 4Ω/8Ω load), up to 55w with increased supply voltage and transformer power;
Continuous output power: defined by heatsink thermal resistance;
Bandwidth (-3dB): 10Hz – 100kHz;
PCB dimensions: 95 x 78mm.



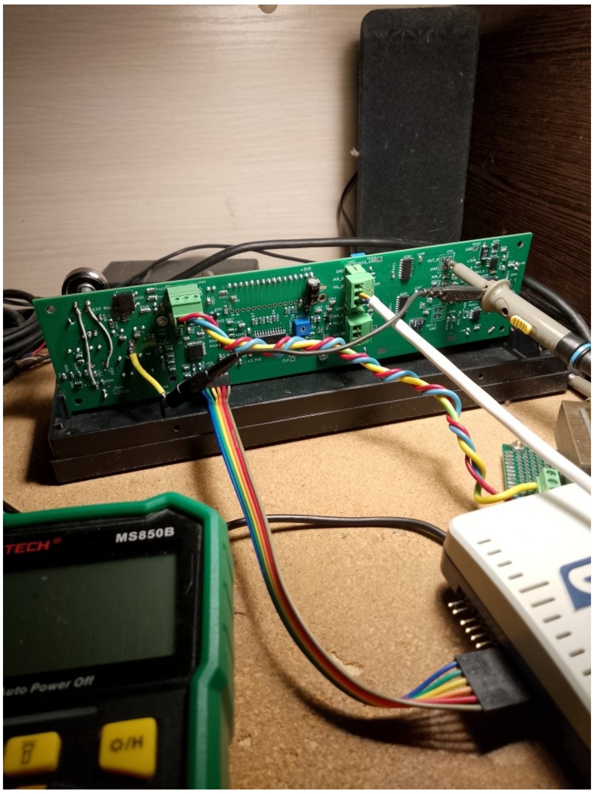


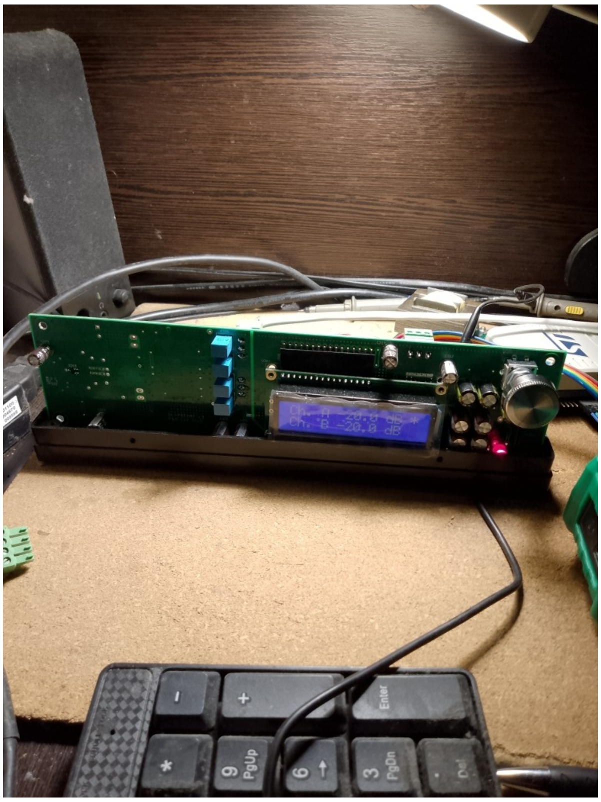
AMP3886 signal stage schematic

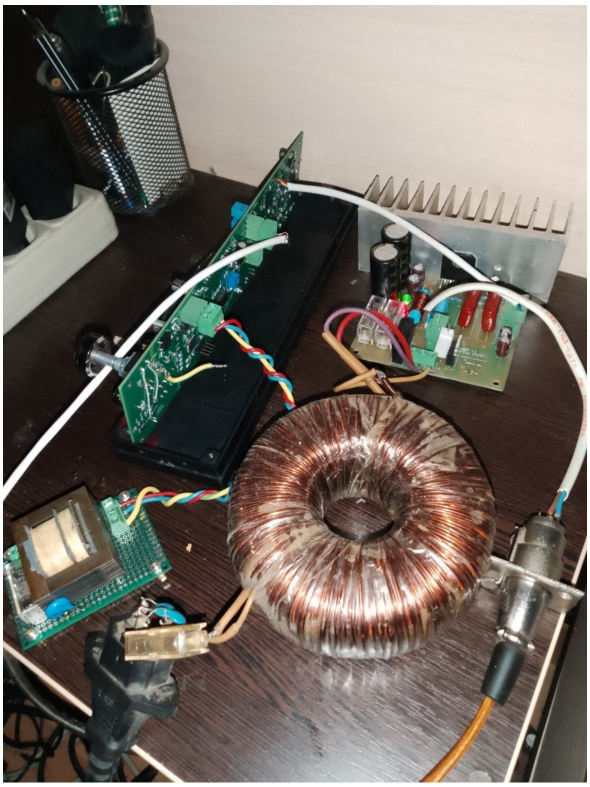


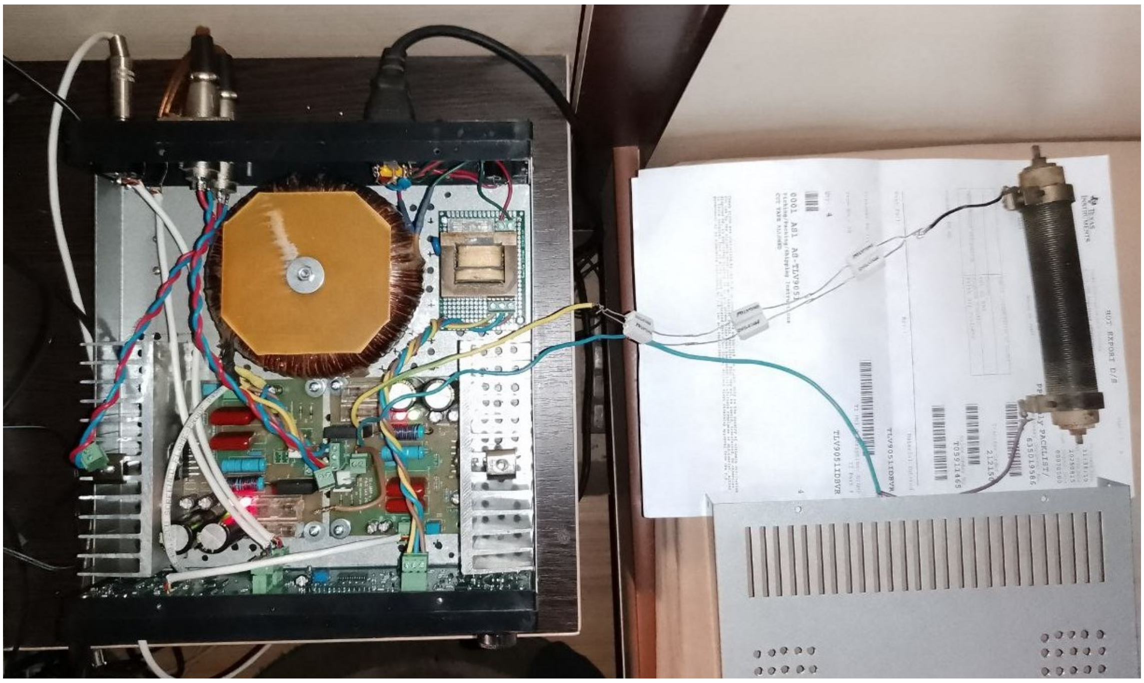
Assembly, startup and testing

Configuration: toroidal transformer 230V to 2x20V with central tap, 150W; heatsinks ~2C/W (~550cm^2); **Supply capacitors – 4700uF LowZ, 150uF LowZ, 1uF X7R, 100nF X7R; LM3886 external components – Cin 180p NP0, Rin 2.2k, Cc 240p NP0, Rf1 33k, Ri 2.2k, Ci 6.6u, Rf2 20k, Cf 27p.** Measured amplifier outputs offset voltage: -1.2mV, -2.0mV.









Caution: the protection circuit is provided as an optional safeguard. Its operation should be validated before soldering the amplifier IC.

Such a circuit was used in a Soviet amplifier “Орбита УМ-002”.

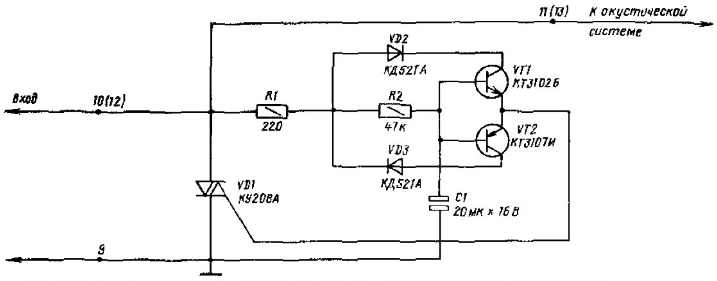
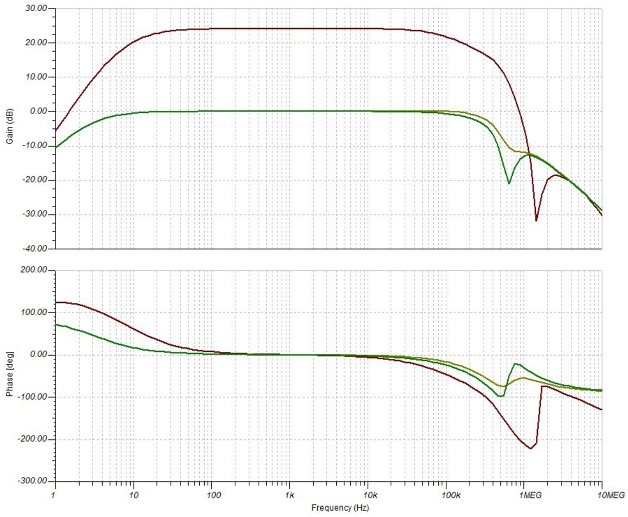
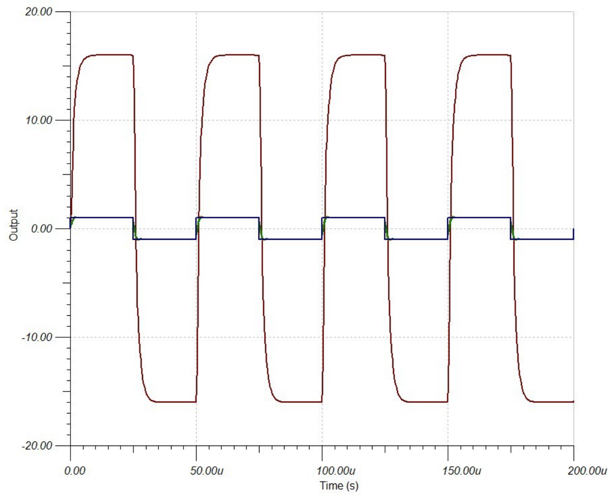


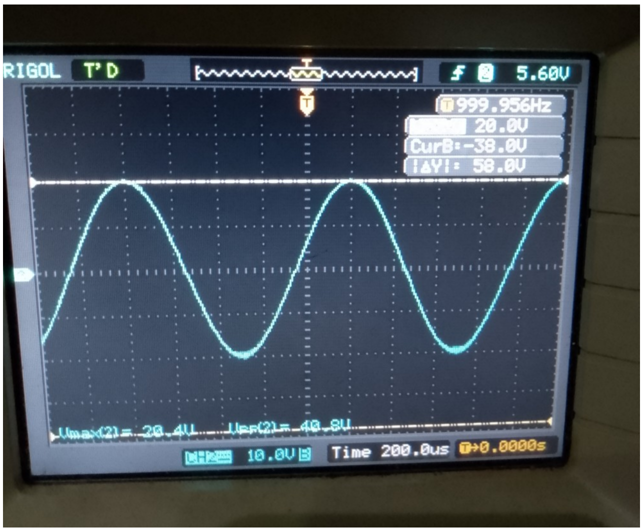
Рис. 16.2. Принципиальная схема устройства защиты громкоговорителя на тиристоре

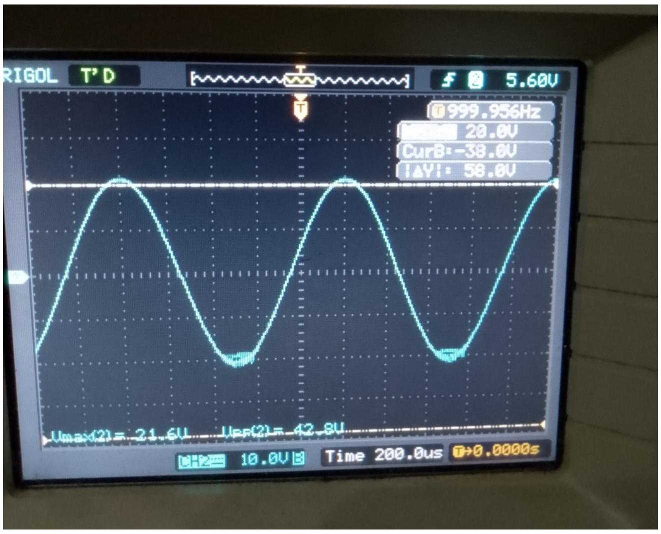
Behaviour at clipping (3.7Ω load).

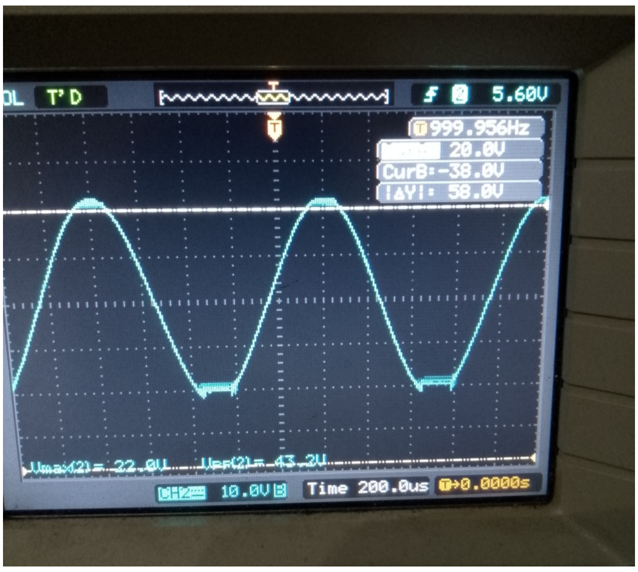
Lowering the gain to 16 and increasing Ri to 2.2k needed to review of the external component values of the LM3886 for stable operation during clipping. Recommended values are: Cin 220p, Rin 2.2k, Cc 220p, Rf1 33k, Ri 2.2k, Ci 9.4u, Rf2 22k, Cf 33p.

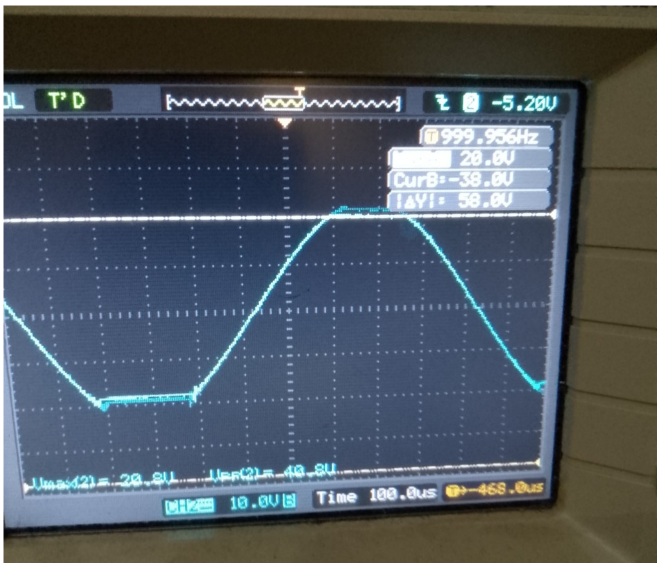




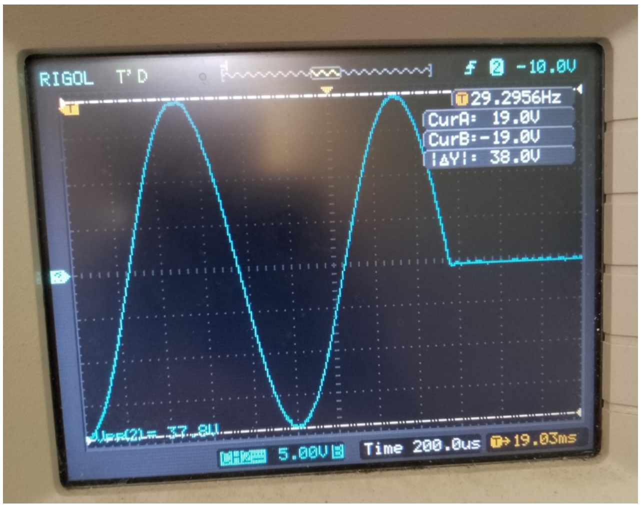


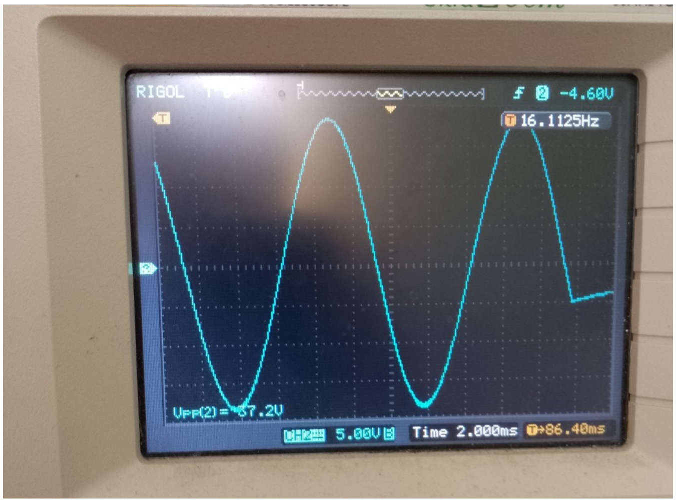




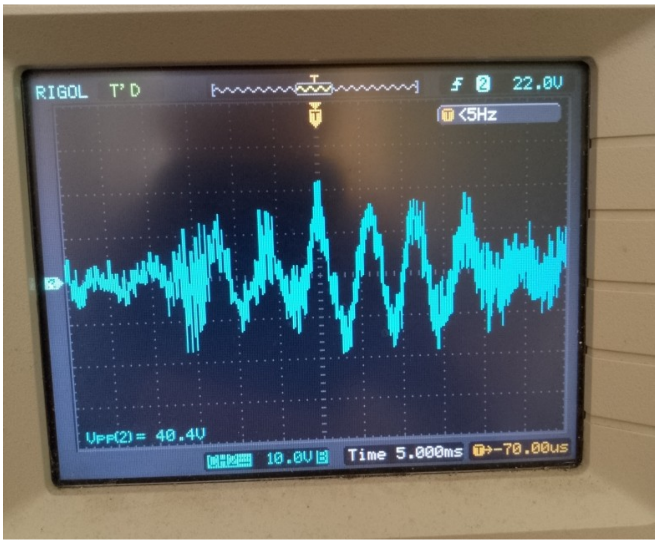


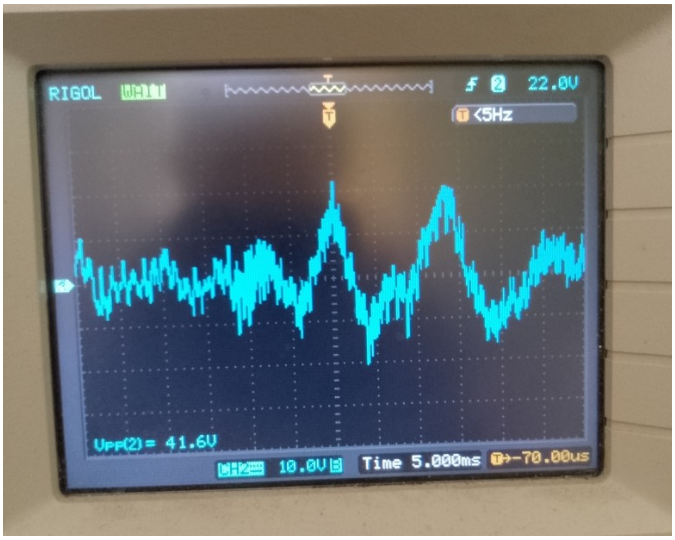
Output power test (3.7Ω load, 1kHz burst 20ms, 100Hz burst 100ms).





Musical signal test (3.7Ω load, PC audio output level 100% - 1V RMS when sinusoidal signal, volume control at 0dB).





After 30 minutes of testing with a high density music signal (low crest factor) in a closed enclosure, ambient temperature 25C, the measured temperatures were:

- heatsink near the IC 75C;

- screw mounting the IC to the heatsink 80C.

In this mode, the amplifier already rarely clips the output signal at about 20V depending on the shape and power of the signal.

In the same operating mode, when driving a 4-ohm loudspeaker system, the temperatures is 10 degrees lower.

The amplifier is subjectively quiet: no audible mains hum or broadband noise at 50 cm from a 85dBw/m midbass driver in a quiet environment (volume at 0dB, input unconnected or PC output in sleep mode). With an active PC audio output, broadband noise becomes noticeable; at a volume setting of −10dB, the noise is barely distinguishable from the unconnected-input condition.

A working prototype (v. 1) was built and evaluated. The design met the intended specifications and no functional issues were observed.

The hardware released (v. 2) is a minor optimized revision of the prototype with no conceptual or architectural changes. A list of significant modifications is provided below.

Change log of preamplifier v.2.0 - 2.6:

1. Added resistor from the buffer op-amp non-inverting input to ground to compensate offset for op-amps with relatively high input bias currents;

2. Added MCU I/O for the 1-Wire interface, 3 output lines for control and 1 dedicated input for a button;

3. Removed the separate 3.3V voltage regulator for the 74LVC8T245 driver;

4. Added TVS diodes to the power input;

5. Added series resistors (10 - 100Ω) in the power lines of the multiplexers;

6. Added capacitor footprints to the multiplexer output (0-220pF for HF input noise and charge injection attenuation);

7. Reverse diode from output to input added to the LM317L and LM337L due to increased output capacitor value.

Change log of power amplifier v.2:

1. Added MLC capacitor footprints to the protection circuitry as alternative for electrolytic;

2. Pin 7 connected to SGND instead of connection to PGND.

Alternative components:

CD74HCT4052 – SOIC-16 CD74HC4052, 74HCT4052, 74HC4052, MC74HC4052A, MAX4052, MAX4582;

OPA1678IDR – OPA1688IDR, RC4580IDR, MC33078DR or any other with acceptable noise and distortion level (preferably with a lower overall output offset voltage);

TLV1117-50IDC – LM1117MPX-5.0, 5V SOT- 223 LD1117, NCP1117;

TPS60403DBV – TPS60400DBV, TPS60401DBV, TPS60402DBV. Not needed if LCD offset voltage is positive.

SN74LVC8T245DW – SN74LVCC3245ADW;

SN74LVC2G17DBV – SN74LVC2G14DBV, 74HC2G17GV, 74HC2G14GV;

6800uF 35V 105C - 4700uF 35V 105C Low impedance.

Sources:

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